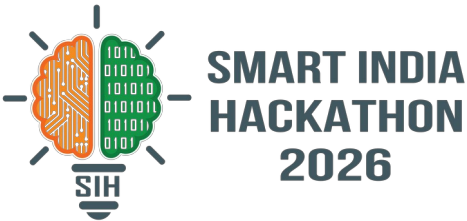


SMART INDIA HACKATHON 2026



RASTA AI

Route Intelligence for Essential-Supply Logistics in North Eastern India

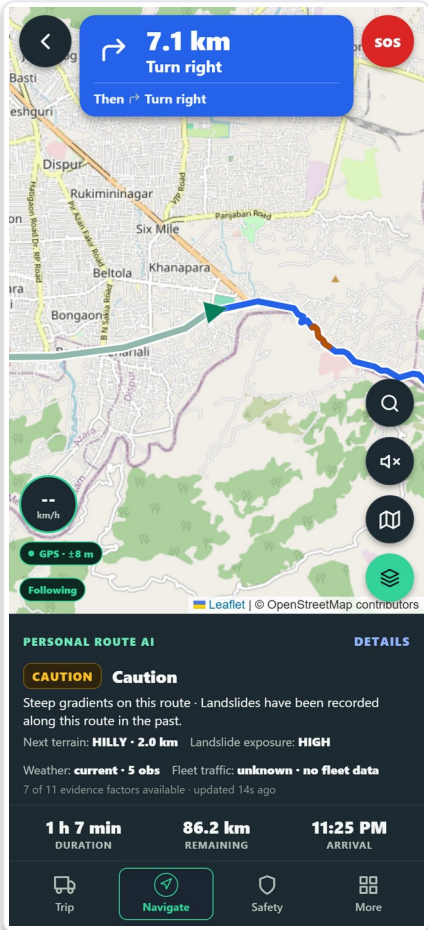
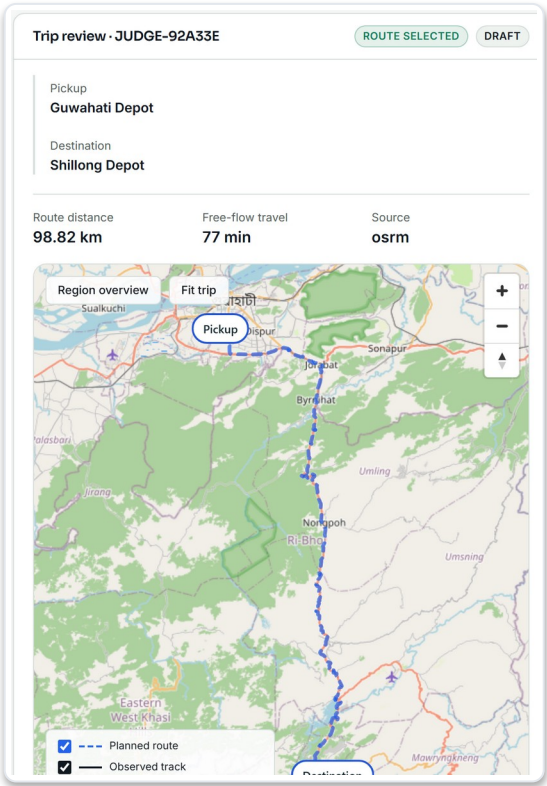
SHORTEST IS NOT ALWAYS USABLE.

“Can this essential-supply truck reliably use this corridor now?”

PROBLEM STATEMENT ID	SIH26002
THEME	Smart Automation
PS CATEGORY	Software
TEAM ID	17 (internal group no.)
TEAM NAME	NER-AI LOGISTICS

AI-Based Smart Logistics and Accessibility Intelligence Platform for North Eastern Region (NER)

LIVE WORKING SYSTEM



Manager route review · Driver navigation — real screens, real road

PROPOSED SOLUTION

Not the shortest road. **The road that is usable now.**

NER corridors close without notice: monsoon rain, landslides, floods, official warnings, single-lane hill roads, no phone signal. A shortest-path router still sends the truck.

A NORMAL ROUTER

Origin → Shortest / fastest road → Truck

No governance. No answer when data is missing.

RASTA AI

Terrain

Weather

Landslide exposure

Flood context

Official warnings

Fleet traffic

Evidence freshness

CONTINUE

CAUTION

HOLD

REROUTE

Manager governance → driver navigation, 23 languages, offline package

UNKNOWN ≠ SAFE

Missing or stale evidence is marked UNKNOWN, never counted as safe, and the route goes to an authorised reviewer.

MANAGER · CHECK CONDITIONS

REQUIRES REVIEW

Hazard evidence is incomplete or elevated. An authorised reviewer must review this route before selection. Refresh conditions after review.

Landslide data is not available - no source is connected. · Steep gradients on this route · Landslides have been recorded along this route in the past. · The landslide inventory ends in 2017 - recent years are not covered. · River levels near this route are close to their recent average. · No official disaster alert names a district on this route right now. · No fleet truck has driven this road in the last fifteen minutes, so traffic is unknown.

Unavailable: Fuel range, Landslide, Road quality, Truck restrictions

Evidence coverage · Weather AVAILABLE · Terrain AVAILABLE · Warnings AVAILABLE · Traffic UNKNOWN

TERRAIN

0.5 KM STEEP

49-1436 m · 1975 m climb · steepest 10.3%
flat 61.8 km · rolling 29.6 km · hilly 7.0 km · steep 0.5 km
Copernicus DEM GLO-90 via Open-Meteo + SRTM 30 m via OpenTopoData · 100% of 198 samples answered

LANDSLIDE HISTORY

HIGH EXPOSURE

22 recorded landslides within 5 km of the road, nearest 0.4 km
6 more nearby placed too imprecisely to count · HIGH is three or more; MODERATE is one or two
NASA Global Landslide Catalog · inventory 2007-2017 — aged, recent years not covered

RIVER LEVELS

NORMAL

Discharge at 5 river cells along the corridor, highest 1.0x its own 30-day mean · GloFAS via Open-Meteo, 2026-09-14. A level, not a flood forecast and not a road-closure claim.

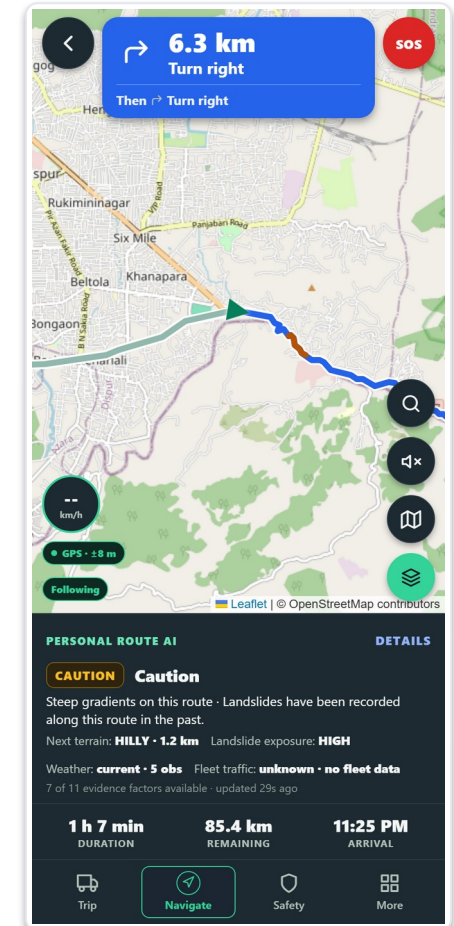
OFFICIAL ALERTS

NONE ON CORRIDOR

NDMA SACHET CAP feed, 99 alerts nationwide at 10:13:54 PM · placed by district name (Dispur, East Khasi Hills, Kamrup Metropolitan, Myllem, Ri-Bhoi, Umiling, Umsning) · 4 more active elsewhere in the corridor states, not placeable on this road.

FLEET TRAFFIC

UNKNOWN



Manager evidence · Driver route AI — the same decision, both ends

VERIFIED DATA

WEATHER

Open-Meteo · MET
Norway

TERRAIN

Copernicus DEM ·
OpenTopoData

WARNINGS

NDMA SACHET
(CAP)

FLOOD

GloFAS

LANDSLIDE

NASA GLC 2007–
17

TRAFFIC

fleet GPS fixes

ROUTE-SPECIFIC EVIDENCE · FastAPI backend

every factor sampled along the selected road · PostgreSQL + PostGIS (Supabase) · OSRM routes, turn steps

DETERMINISTIC SAFETY POLICY

11 factors · reason codes · UNKNOWN ≠ SAFE · reviewer authorisation when hazard data is missing

CONTINUE

CAUTION

HOLD

REROUTE

MANAGER REVIEW

DRIVER NAV

60 s WATCH

REASSESS

Re-scored every 60 s along the road ahead; a reroute is proposed to the manager, never applied silently.

DETERMINISTIC SAFETY ENGINE

makes every operational decision

GEMINI / OPENROUTER (LLM)

explains the decision in the driver's language — never decides

EXPERIMENTAL ML

research channel only; held outside routing by the safety gate

STACK · CLIENTS · GOVERNANCE

CLIENTS Manager console — React + TypeScript, manager accounts only · Driver app — React Native / Expo, Android; one login, server-decided role, offline trip package**STACK** FastAPI · PostgreSQL / PostGIS (Supabase) · React · React Native / Expo · OSRM · OpenStreetMap · Gemini

GOVERNANCE CHAIN



Diagnostics reports every provider's real state — healthy, degraded, fallback active or unknown — so a judge can see exactly what a decision rested on.

Not a concept. A working end-to-end prototype.

CERTIFIED LIFECYCLE

PLAN

ACCEPT

OFF-ROUTE

REVIEW

TRUCK VERIFY

MANAGER REROUTE
APPROVAL

DISPATCH

NAVIGATE

DELIVERY

12/12 physical judge-flow steps

9/9 physical role-flow steps

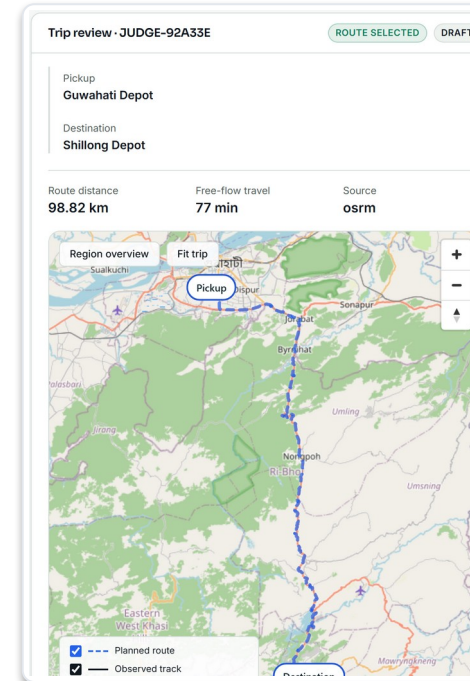
1138+ backend · 621+ driver · 170+ manager tests

PHYSICAL ANDROID · CERTIFIED

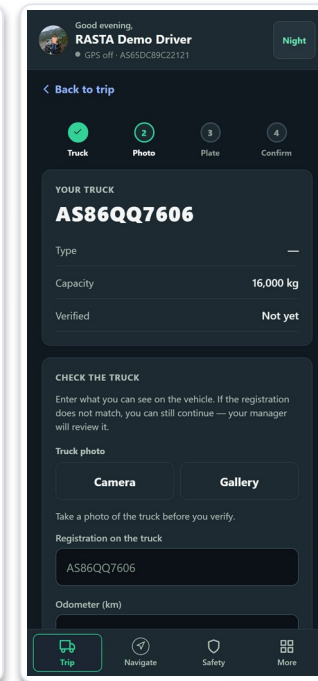
Canonical demo Guwahati → Shillong · ≈ 98.8 km real route

KNOWN LIMITS → MITIGATION Render cold start → warm /health before the demo · provider rate limits → cached evidence + explicit UNKNOWN · background push needs Firebase config → in-app alerts already work · experimental ML → outside routing until the gate passes

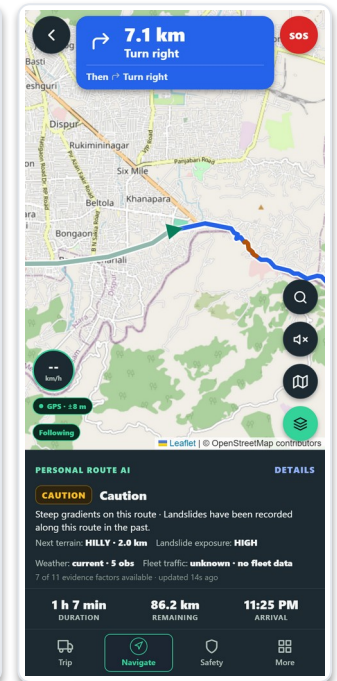
REAL SCREENS FROM THE CERTIFIED RUN



Manager · route review



Driver · truck check



Driver · navigation

“Can this truck reliably use this corridor now?”

FOOD · MEDICINE · FUEL · RELIEF SUPPLIES

essential logistics for hill communities, decided on evidence

FLEET MANAGERS

Route usability and uncertainty visible before dispatch; auditable decisions.

DRIVERS

Safer navigation, danger context, emergency support, 23 languages.

REMOTE COMMUNITIES

More resilient access to food, medicine, fuel and relief supplies.

GOVERNMENT / OPERATIONS

Auditable route evidence and disruption visibility per corridor.

23 selectable languages, explicit fallback status

11 route factors in every decision

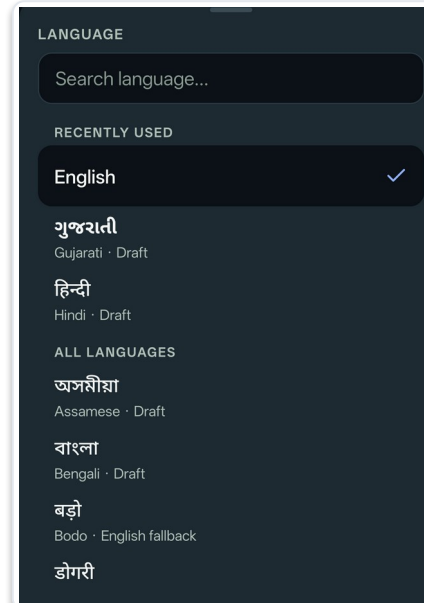
60 s route-ahead monitor while driving

12/12 judge-flow steps certified on a physical phone

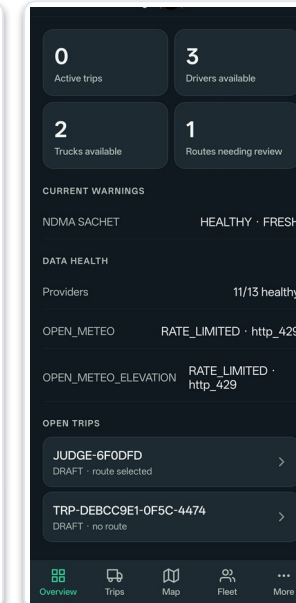
100 % of reroutes reviewed by a human

0 unknown factors counted as safe

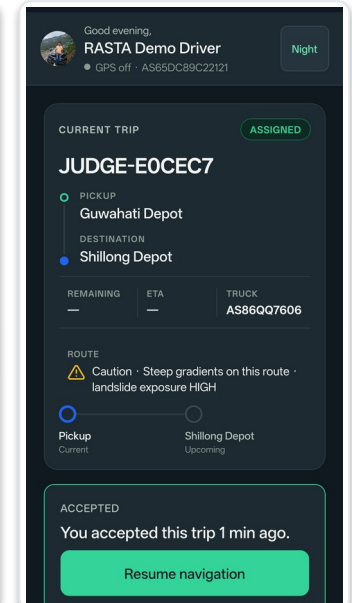
REAL SCREENS · APK 1.0.18



23 languages



Manager · mobile



Driver · assigned trip

HUMAN GOVERNANCE — AI HAS NO UNCONTROLLED AUTHORITY



EXPERIMENTAL LANDSLIDE RESEARCH

95 %

RECALL

on the NER geographic holdout — 154 catalogued events, no NER row used in training

SAFETY GATE
NOT DEPLOYED**FPR 50 %** too high for production deployment

TRAIN

HELD-OUT NER TEST

USEFUL SIGNAL FOUND

FALSE POSITIVES TOO HIGH

SAFETY GATE

NOT DEPLOYED

VALIDATION BEFORE AUTOMATION. Strong recall was not enough. Held-out testing exposed the false-positive rate, so the gate keeps the model out of routing.

Precision 0.32 · F1 0.48 · PR-AUC 0.55 · ROC-AUC 0.83 · Brier 0.12 · calibrated logistic regression on NASA GLC + ERA5-Land rain + DEM slope · docs/HAZARD_ERROR_ANALYSIS.md

RESEARCH / EVIDENCE SOURCES BEHIND THE ROUTE DECISION

ROUTING / PLACES

OSRM · OpenStreetMap · Nominatim

real road geometry, alternatives and turn steps; place search on the corridor

WEATHER / TERRAIN

Open-Meteo · MET Norway · Copernicus DEM · OpenTopoData

forecast sampled along the road; climb, gradient and steep segments (GLO-90)

OFFICIAL / FLOOD

NDMA SACHET (CAP) · Copernicus GloFAS

official warnings polled with freshness; river discharge against its 30-day mean

LANDSLIDE HISTORY

NASA Global Landslide Catalog

historical exposure; inventory 2007–2017, ≤5 km location-accuracy filter

AI ASSISTANCE

Google Gemini

explains the decision in the driver's language — never safety authority

PROBLEM STATEMENT

MDoNER · SIH26002

AI-based smart logistics and accessibility intelligence for the NER

REFERENCES gpm.nasa.gov/landslides (NASA GLC) · sachet.ndma.gov.in · global-flood.emergency.copernicus.eu (GloFAS) · open-meteo.com · api.met.no · opentopodata.org · Copernicus DEM · project-osrm.org · openstreetmap.org · github.com/nxtlucifer/ner-ai-logistics (repository, tests, model registry)